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ISBN 9781284296327
CHAPTER 3 Epidemiologic Measurements Used to Describe Disease Occurrence
Measures of Natality and Mortality Linked to Natality

Measures of Natality and Mortality Linked to Natality

Measures of natality include the **crude birth rate** and the fertility rate. Additionally, statisticians compute measures that describe mortality linked to natality. These indices are the maternal mortality rate, fetal mortality rate, and infant mortality rate. This section covers several measures that pertain to the number of births in a population (birth rate) and the fertility of people of childbearing age (fertility rate). Note that by statistical convention, one definition of the childbearing age is 15 to 44 years of age. Related to giving birth is maternal mortality, which occurs during a small number of births in this country. Another fatal outcome is death of the fetus during gestation, which is called fetal mortality. Still another measure tracks death of the newborn during the first year of life, which is called infant mortality. Consult **Table 3.5** for measures presented in this section as well as some of their applications.

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TABLE 3.5 Examples of Measures of Natality and Mortality Linked to Natality

Measure	How It Is Used
Maternal mortality rate	Reflects health disparities such as healthcare access
Infant mortality rate	For international comparisons to identify countries with high rates
Fetal death rate/late fetal death rate	Measures risk of death of the fetus
General fertility rate	Compares populations and subgroups regarding their fertility
Crude birth rate	To project population changes

Maternal Mortality

Maternal mortality encompasses maternal deaths that result from causes associated with pregnancy. The **maternal mortality rate** is the number of maternal deaths ascribed to childbirth divided by the number of live births times 100,000 live births during a year. Among the factors related to maternal mortality are age and race. In 2020, the maternal mortality rate was 23.8 deaths per 100,000 live births (861 total deaths in 2020) in the United States, where the maternal mortality rate has been increasing slightly in recent years. In the United States, large differences are observed in maternal

mortality rates by race, with the rates in 2020 for non-Hispanic Black women being more than double the rates for non-Hispanic White women (55.3 versus 19.1 deaths per 100,000 live births, respectively)⁹ Maternal mortality rates in the United States are among the highest compared to other high-income countries. The rate has more than doubled since 1999. The vast differences between maternal mortality rates in Black women compared to White women are a reflection of social determinants of health including inadequate access to prenatal care and the effects of systemic racism.¹⁰

Maternal Mortality Rate

$$= \frac{\text{Number of deaths assigned to a cause related to childbirth}}{\text{Number of live births}} \times 100,000$$

- Time period needs to be specified (usually one year).
- Note: Live births include multiple births

Infant Mortality Rate

The **infant mortality rate** is defined as the number of infant deaths among infants age 0 to 365 days during a year divided by the number of live births during the same year (expressed as the rate per 1,000 live births). Refer to the text box.

The **neonatal mortality rate** is the number of infant deaths under 28 days of age divided by the number of live births during a year.

Infant Mortality Rate

$$= \frac{\text{Number of deaths among infants age 0–365 days}}{\text{Number of live births}} \times 1,000$$

- Time period needs to be specified (usually one year).
- *Calculation example:*
In the United States during 2020,
 - 19,582 deaths among infants younger than 1 year.
 - 3,613,647 live births.
 Infant mortality rate

$$= \frac{19,582}{3,613,647} \times 1,000 = 5.42 \text{ per 1,000 live births}$$

From 1995 to 2019, the infant mortality rate in the United States declined by 26% (7.57 versus 5.58). Infant mortality is related to inadequate prenatal care and poor environmental conditions. There is substantial variation in infant mortality across states. (See **Figure 3.13.**)

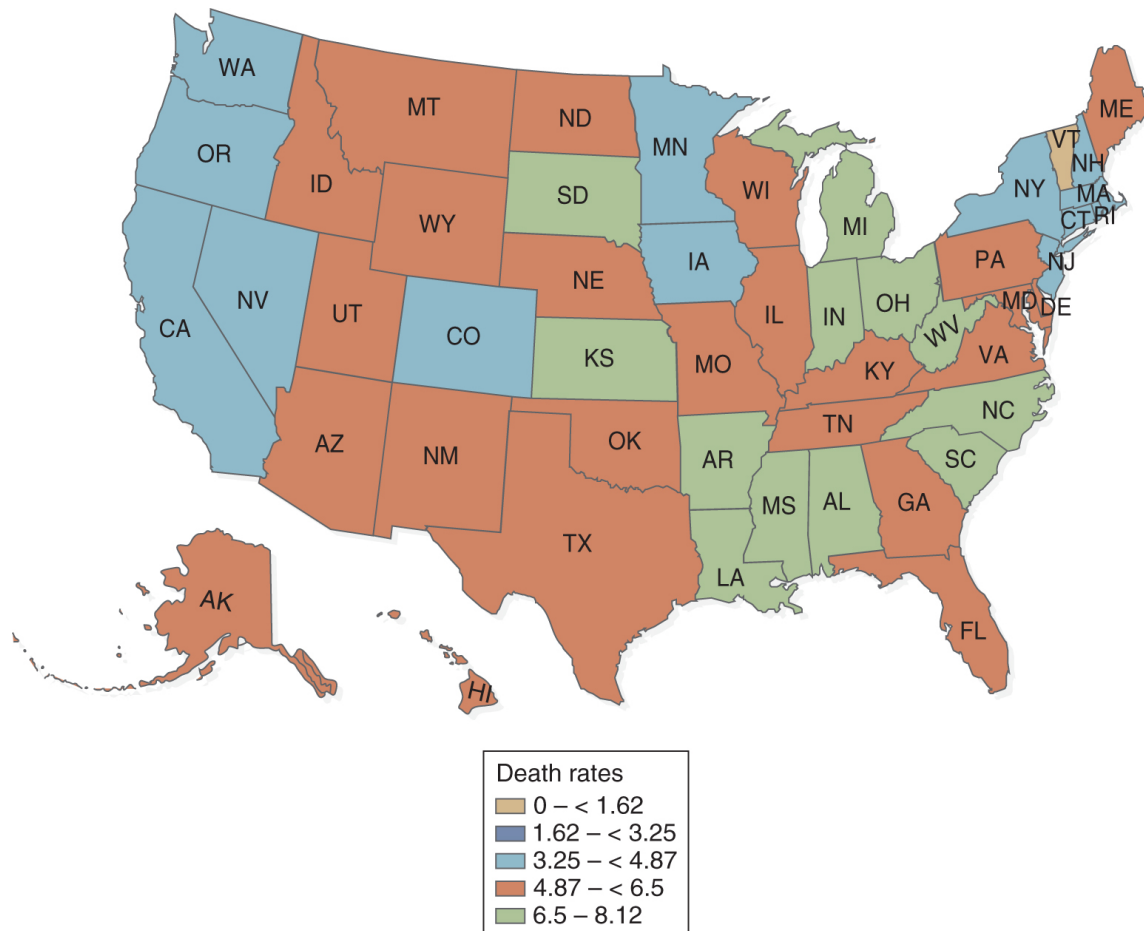


Figure 3.13 Infant mortality rate, by state: United States, 2020

Ely DM, Driscoll AK. *Infant mortality in the United States, 2019: Data from the period linked birth/infant death file*. National Vital Statistics Reports; vol. 70 no. 14. Hyattsville, MD: National Center for Health Statistics; 2021.

Description

Fetal Mortality

Fetal mortality is defined as the death of the fetus when it is in the uterus and before it has been delivered. Two measures of fetal mortality are the